A.I Fundamentals

Module 3 Study Guide

Neural Networks & Deep Learning

This guide covers every lesson of the module, using only words and figures from the lessons themselves. Work through it after watching the lessons, then use the checklist at the end to test yourself.

Lesson 1: Inside a Neural Network

In this lesson: Architecture of Neural Networks; Understanding Weights and Importance; Work

Key points

- Input layer gathers raw facts
- Hidden layers weigh importance with dials
- Activation function triggers the output prediction

Examples from the lesson

- We are kicking off Module 3, "Neural Networks & Deep Learning," with Lesson 1: "Inside a Neural Network."
- You do not just look at one factor; your brain processes several signals all at once.
- At its core, a neural network is built out of three main parts: the input layer, the hidden layers, and the output layer.
- Imagine you want to buy a ticket for a big Carnival fete in Chaguaramas costing TT$ 800.
- You input two factors into your personal financial network: your available discretionary cash after paying rent, which is TT$ 1,000, and your monthly credit union savings contribution, which sits at TT$ 300.
- The network applies a heavy positive weight of 0.8 to your TT$ 1,000 in free cash because having liquid money right now is very important.
- That gives you a weighted signal of TT$ 800.
- However, the network puts a negative weight of minus 0.5 on taking money away from your TT$ 300 credit union savings, resulting in a penalty signal of minus TT$ 150.
- It takes your TT$ 800 positive signal from free cash and subtracts the TT$ 150 penalty from disrupting your savings.
- This leaves you with a combined raw score of TT$ 650.
- It takes that raw score of TT$ 650 and passes it through a rule: if the combined score passes a threshold of TT$ 500, the switch flips to ON.
- Because TT$ 650 is greater than TT$ 500, the activation function fires a green light to the output layer, predicting a "Safe Purchase."
- A neural network turns raw data into reliable predictions using three connected steps: the input layer gathers your facts, the hidden layers use weighted dials to measure importance,

and the activation function acts as a gatekeeper to push the final answer through the output layer.

- In our next lesson, Lesson 2: "Deep Learning in Practice," we will see how stacking many hidden layers together allows deep learning systems to handle complex real-world financial data across Trinidad & Tobago.

Lesson 2: Deep Learning in Practice

In this lesson: Traditional vs Deep Learning; Document Processing Economics; Computer Visior

Key points

- Deep learning automates feature extraction from unstructured data
- It drives major cost savings in high-volume document workflows
- Visual and language AI transform real-world operations

Examples from the lesson

- I am your instructor, guiding you through course "A.I Fundamentals", Module 3 "Neural Networks & Deep Learning".
- Today, we are diving into Lesson 2: "Deep Learning in Practice".
- Let's compare the cost structure of processing 1,000 document packages per month.
- Under traditional manual and rule-based processing, document intake costs TT$ 45,000, data entry labor costs TT$ 35,000, and error reconciliation costs TT$ 20,000, bringing total monthly intake expenses to TT$ 100,000.
- The deep learning software subscription costs TT$ 15,000 per month, cloud compute costs TT$ 5,000, and human oversight for edge cases costs TT$ 10,000, bringing total monthly expenses down to TT$ 30,000.
- By calculating the difference, the financial institution achieves monthly savings of TT$ 70,000.

Lesson 3: Generative AI and Large Language Models

In this lesson: How LLMs Predict Text; Understanding AI Hallucinations; Verifying AI Financial I

Key points

- Generative AI predicts probable words rather than looking up facts, leading to hallucinations
- A side-hustle budget example revealed a TT$700 variance between AI predictions and verified local costs
- Verify all AI financial outputs against local market prices and official regulations before acting

Examples from the lesson

- I am your instructor, and welcome to "A.I Fundamentals", Module 3: "Neural Networks & Deep Learning", Lesson 3: "Generative AI and Large Language Models".
- The AI outputs: raw ingredients costing TT$2,200, stall rental and transport costing TT$800, and marketing expenses costing TT$500, bringing your predicted monthly expenses to TT$3,500.
- The AI then claims your projected gross income will be TT$5,500 based on standard street festival margins.
- If you subtract the predicted monthly expenses of TT$3,500 from the projected gross income of TT$5,500, the AI claims your projected net profit will be TT$2,000.
- To verify this AI output, you must run every line item through a three-step fact-checking process.
- For instance, after checking current market rates, your actual raw ingredients come out to TT$2,600, stall rental and maxi transport cost TT$1,000, and local social media ads cost TT$600, making your actual monthly expenses TT$4,200.
- Subtracting your actual monthly expenses of TT$4,200 from the realistic gross income of TT$5,500 leaves you with an actual net profit of TT$1,300, not TT$2,000.
- That TT$700 variance between AI prediction and ground reality represents the cost of unverified hallucinations.
- We worked through a side-hustle budget example showing how an AI prediction of TT$2,000 net profit differed from the verified ground reality of TT$1,300 net profit, saving TT$700 in hidden errors.
- That wraps up Module 3 on Neural Networks & Deep Learning!
- Join us in Module 4, Lesson 1: "Prompt Engineering for Everyday Finance", where we will learn how to write precise prompts to get accurate, actionable financial insights from AI tools.

Recap checklist — can you say each one?

Lesson 1: Inside a Neural Network

- ☐ Input layer gathers raw facts
- ☐ Hidden layers weigh importance with dials
- ☐ Activation function triggers the output prediction

Lesson 2: Deep Learning in Practice

- ☐ Deep learning automates feature extraction from unstructured data
- ☐ It drives major cost savings in high-volume document workflows
- ☐ Visual and language AI transform real-world operations

Lesson 3: Generative AI and Large Language Models

- ☐ Generative AI predicts probable words rather than looking up facts, leading to hallucinations
- ☐ A side-hustle budget example revealed a TT$700 variance between AI predictions and verified local costs
- ☐ Verify all AI financial outputs against local market prices and official regulations before acting